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Unit – 4 : Session – 2 : SLO – 1 | Pure Functions with Examples

■ I. Match the Following

1. `map()` — Applies a function to each element. Example: `list(map(str.upper, ["a", "b"])) → ["A", "B"]`.
2. `filter()` — Selects elements based on a condition. Example: `list(filter(lambda x: x > 0, [-1, 2, -3, 4])) → [2, 4]`.
3. `reduce()` — Accumulates values to a single result. Example: `reduce(lambda a, b: a + b, [1, 2, 3, 4]) → 10`.
4. Pure Function — Same output for the same input, with no side effects. Example: `def add(a, b): return a + b`.

■ II. Key Notes

- A pure function never mutates its arguments, global variables, or performs I/O — its result depends only on its inputs.
- `map()` and `filter()` return lazy iterators in Python 3; wrap them in `list()` to materialize the results.
- `reduce()` lives in the `functools` module: `from functools import reduce`.